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The Structure of Information Networks	**Homework 2**
CS/INFO 6850 Fall 2024	**Due 6pm, Wednesday, October 2, 2024**

(1)

(a) Let E_j be the event that no sensor lands in the interval $[j-1, j]$, for $j = 1, 2, \ldots, n$. Since each sensor is placed uniformly at random over the interval $[0, n]$, the probability that a given sensor does not land in $[j-1, j]$ is $1 - \frac{1}{n}$

Thus, the probability that none of the k sensors land in the interval $[j-1, j]$ is simply the product of the probabilities for each sensor is

$$\Pr[E_j] = \left(1 - \frac{1}{n}\right)^k$$

For large n, we can use the approximation $\left(1 - \frac{1}{n}\right)^k \approx e^{-k/n}$. Therefore, the probability that no sensor lands in the interval $[j-1, j]$ is empty is approximately as:

$$\Pr[E_j] \approx e^{-k/n}$$

(b) If none of the events E_j occurs, it means that in each interval $[j-1, j]$, there is at least one sensor. Each sensor can communicate with other sensors within a distance of 2 units, and this means that the sensors in adjacent intervals can always have a path connecting them. Therefore, the graph G will be connected.

(c) Now we calculate the probability that at least one interval is empty, which would imply that the graph is disconnected. Using the Union Bound to estimate this probability:

$$\Pr\left[\bigcup_{j=1}^{n} E_j\right] \leq \sum_{j=1}^{n} \Pr[E_j]$$

Each interval $[j-1, j]$ has the same probability of being empty, $\Pr\left[\bigcup_{j=1}^{n} E_j\right] \leq n \cdot \Pr[E_j]$

For large n, we know that $\Pr[E_j] = \left(1 - \frac{1}{n}\right)^k \approx e^{-k/n}$. Thus, $\Pr\left[\bigcup_{j=1}^{n} E_j\right] \leq ne^{-k/n}$

Then, substituting $k = 2n \ln n$ into the inequality: $\Pr\left[\bigcup_{j=1}^{n} E_j\right] \leq ne^{-2 \ln n/n}$

Then, simplifying: $ne^{-2 \ln n/n} = n \cdot e^{-2 \ln n} = n \cdot \frac{1}{n^2} = \frac{1}{n}$

Thus, the probability that any interval is empty is at most $\frac{1}{n}$. As $n \to \infty$, this probability tends to 0. Therefore, with $k = 2n \ln n$ sensors, the graph G is connected with high probability.

(2)

(a) Let Z_j represent the in-degree of node v_j, i.e., the number of incoming links to node v_j. We can express Z_j as a sum of indicator random variables $X_{k,j}$, where $X_{k,j} = 1$ if node v_k (for $k > j$) links to node v_j, and $X_{k,j} = 0$ otherwise. Thus,

$$Z_j = \sum_{k=j+1}^{n} X_{k,j}$$

Each $X_{k,j}$ is a Bernoulli random variable with probability $\frac{1}{k-1}$, since each node v_k chooses its link destination uniformly at random from the $k-1$ nodes that already existed.

(b) Applying the linearity of expectation, we can compute the expected in-degree $E[Z_j]$:

$$E[Z_j] = \sum_{k=j+1}^{n} E[X_{k,j}] = \sum_{k=j+1}^{n} \frac{1}{k-1}$$

Since the harmonic number H_k is defined as $H_k = \sum_{i=1}^{k} \frac{1}{i}$:

$$\sum_{k=j+1}^{n} \frac{1}{k-1} = \sum_{i=j}^{n-1} \frac{1}{i}$$

Thus, the expected in-degree of node v_j can be written as the difference between two harmonic numbers:

$$E[Z_j] = H_{n-1} - H_{j-1}$$

(c) Let X_j be a random variable that is equal to 1 if node v_j has no incoming links, and 0 otherwise. The probability that a node v_k does not link to v_j is $1 - \frac{1}{k-1}$. Therefore, the probability that none of the nodes v_k (for $k > j$) link to v_j is the product:

$$E[X_j] = \prod_{k=j+1}^{n} \left(1 - \frac{1}{k-1}\right)$$

For large n, this product approximates e^{-1} since it converges to e^{-1}. Thus, the expected value of X_j:

$$E[X_j] \approx \frac{1}{e}$$

(d) Using the linearity of expectation, the expected number of nodes with no incoming links is $E\left[\text{Number of nodes with no incoming links}\right] = \sum_{j=1}^{n} E[X_j]$

Since $E[X_j] \approx \frac{1}{e}$, the expected number of nodes with no incoming links is approximately:

$$\sum_{j=1}^{n} E[X_j] \approx n \cdot \frac{1}{e} = \frac{n}{e}$$

$$E\left[\text{Number of nodes with no incoming links}\right] \approx \frac{n}{e}$$

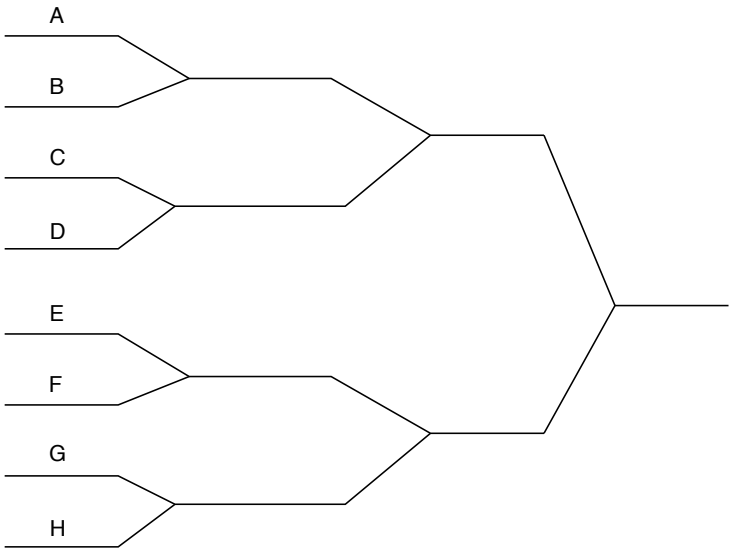


Figure 1: An elimination tournament of the kind described in Question 3, with just the initial contestants, before any matches have been played.

(3)

(a) We are tasked with finding the probability $p(j)$, which represents the probability that at least one of the guesses for round j in a single-elimination tournament is correct.

In this tournament:

- The total number of contestants is $n = 2^k$, where k is the number of rounds.
- After each round, the number of contestants is halved.
- Each match outcome is determined randomly, and the probability of guessing correctly for a single match is $\frac{1}{2}$.

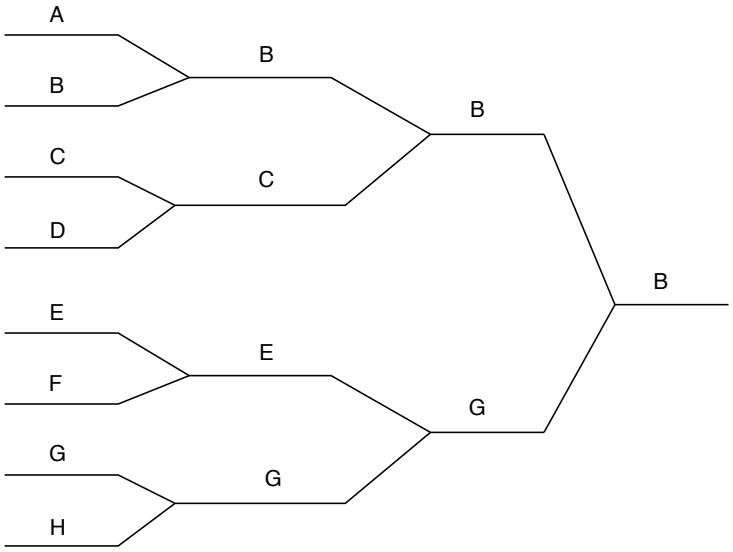


Figure 2: The results of the elimination tournament from Figure 1, after all matches have been played.

In each round of a single-elimination tournament, half the contestants are eliminated. Thus, after j rounds, the number of contestants remaining in the tournament is $\frac{n}{2^j}$ where $n = 2^k$ is the initial number of contestants.

For each match in round j, the probability of guessing correctly for a single match is $\frac{1}{2}$. Since the number of contestants after round j is $\frac{n}{2^j}$, there are $\frac{n}{2^{j+1}}$ matches in that round because each match involves two contestants.

The probability of guessing wrong for all matches is $\frac{1}{2}$. The probability of guessing incorrectly for all matches in round j is:

$$\Pr[\text{all guesses wrong in round j}] = \left(\frac{1}{2}\right)^{\frac{n}{2^{j+1}}} = \frac{1}{2^{n/2^{j+1}}}$$

Then, the probability of guessing at least one match correctly is

$$p(j) = 1 - \Pr[\text{all guesses wrong}]$$

Therefore, the probability $p(j)$ that at least one of the guesses for the contestants in round j is correct is:

$$p(j) = 1 - \frac{1}{2^{n/2^{j+1}}}$$

(b)

From part 3(a), we know that $p(j) = 1 - \frac{1}{2^{n/2^{j+1}}}$

Substituting $j = \alpha \log n$:

$$p(\alpha \log n) = 1 - \frac{1}{2^{n/2^{\alpha \log n+1}}}$$

Simplifying the exponent $2^{\alpha \log n+1} = 2^{\alpha \log n} \cdot 2^1 = 2n^\alpha$

$$p(\alpha \log n) = 1 - \frac{1}{2^{n/2n^\alpha}} = 1 - \frac{1}{2^{n^{1-\alpha}/2}}$$

When $\alpha < 1$, we have $1 - \alpha > 0$, and $n^{1-\alpha} \to \infty$.

$$2^{n^{1-\alpha}/2} \to \infty \quad \text{and} \quad \frac{1}{2^{n^{1-\alpha}/2}} \to 0$$

Thus, $p(\alpha \log n) = 1 - 0 = 1$. The probability of making at least one correct guess tends to 1 as $n \to \infty$ for $\alpha < 1$.

When $\alpha > 1$, we have $1 - \alpha < 0$, and $n^{1-\alpha} \to 0$.

$$2^{n^{1-\alpha}/2} \to 1 \quad \text{and} \quad \frac{1}{2^{n^{1-\alpha}/2}} \to 1$$

Thus, $p(\alpha \log n) = 1 - 1 = 0$. The probability of making at least one correct guess tends to 0 as $n \to \infty$ for $\alpha > 1$.

At $\alpha = 1$:

$$p(1 \log n) = 1 - \frac{1}{2^n} \quad \text{and as} \quad n \to \infty, \quad \frac{1}{2^n} \to 0$$

Thus, $p(1 \log n) = 1$. The probability of making at least one correct guess tends to 1.

There, the critical point is the value $\alpha^8 = 1$ because it separates two distinct behaviors of $p(\alpha \log n)$:

- When $\alpha < 1$, $p(\alpha \log n) \to 1$, the probability of guessing at least one match correctly is very high.
- When $\alpha > 1$, $p(\alpha \log n) \to 0$, the probability of guessing any match correctly becomes very low.